Darshana Opticals (Pvt) Ltd

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Software Requirement Specification (SRS)

Version History

The following table provides a comprehensive record of the revisions made to this Software Requirement Specification, ensuring that all changes are documented for auditability and project tracking as required by the university standards for design projects.

Version No.	Date	Author(s)	Summary of Changes
0.1	2025/11/27	Project Team	Initial requirements gathering and concept definition for the initial University Presentation.
0.2	2026/01/15	Project Team	Refinement of system scope and high-level features for the Project Poster submission.
1.0	2026/03/21	Project Team	Final SRS document; integration of 21-section structure, feasibility studies, and detailed diagrams for final design approval.

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1. Introduction

The transition of Darshana Optical Pvt Ltd from a manual physical-store model to a comprehensive digital platform is addressed in this document. The foundational details of the proposed Optical Shop Management System (OSMS) are provided in the following sections.

2. Purpose

The primary purpose of this Software Requirement Specification (SRS) document is to provide a detailed description of the Optical Shop Management System (OSMS) for Darshana Optical Pvt Ltd. The functionalities required to digitize business operations, including the online store, chatbot, and loyalty programs, are explicitly outlined.

Intended Audience: This document is intended for the following stakeholders:

- Project Managers and Developers: To serve as a definitive technical guide for system behavior and constraints.
- Darshana Optical Management: To establish a shared understanding of the automated solutions provided.
- Quality Assurance Teams: To define the standards against which the system's performance and security are verified.

3. Scope

System Name: The system is identified as the **Optical Shop Management System (OSMS)**.

What the System Will Do: The OSMS is designed as a web-based platform to automate the core business processes of Darshana Optical Pvt Ltd. The following capabilities are provided:

- **Digital Presence:** Eyewear and accessories are showcased in an online store, allowing for digital browsing and purchasing.
- **Appointment Management:** An online booking system for eye examinations is implemented with automated reminders.
- **Inventory Tracking:** Stock levels across all branches are synchronized in real-time to eliminate manual tracking errors.
- **Customer Engagement:** Instant assistance is provided via an AI-driven chatbot, and customer retention is managed through a loyalty program.

Explicit Exclusions: The following areas are explicitly excluded from the system scope:

- **Medical Diagnostic Hardware:** Management of eye test data is performed, but direct interfacing with medical diagnostic hardware is not supported.

- **Physical Delivery:** Order tracking is included; however, the actual physical delivery and fleet management are handled by external third-party logistics.
- **Financial Auditing:** Sales and discounts are processed, but full corporate financial auditing or tax filing modules are not included in this system.

4. Definitions, Acronyms and Abbreviations

All domain-specific terms, technical acronyms, and abbreviations used throughout this Software Requirement Specification (SRS) document are defined in the following table to ensure a common understanding among all stakeholders.

Term	**Definition**
OSMS	**Optical Shop Management System** – The proposed digital platform designed for Darshana Optical Pvt Ltd.
SRS	**Software Requirement Specification** – The formal document that describes the features and behavior of the software system.
MERN	**MongoDB, Express.js, React, Node.js** – The full-stack JavaScript framework selected for the development of this system.
JWT	**JSON Web Token** – A compact, URL-safe means of representing claims to be transferred between two parties, used for secure user authentication.
SKU	**Stock Keeping Unit** – A unique alphanumeric code assigned to inventory items to track stock levels and variants.
AI	**Artificial Intelligence** – The technology utilized in the system’s chatbot to provide automated customer assistance.
UI/UX	**User Interface / User Experience** – The visual design and the overall experience provided to the end-users of the OSMS.

HTTP/HTTPS	**Hypertext Transfer Protocol (Secure)** – The underlying protocol used for data communication over the web.
API	**Application Programming Interface** – A set of defined rules that allow different software entities to communicate with each other.
ERD	**Entity Relationship Diagram** – A visual representation of the database structure and the relationships between data entities.

Table 01

5. Document Overview

The organization of the remainder of this Software Requirement Specification (SRS) document is described as follows:

- Section 6 (Project Scope): A detailed boundary of the project, including the specific business goals of Darshana Optical Pvt Ltd, is provided.
- Section 7 (Stakeholders): The key individuals and groups influenced by the system, including administrators, staff, and customers, are identified.
- Section 8 (Functional Requirements): The core services provided by the system, documented in a tabular format with unique IDs and priority levels, are explicitly defined.
- Section 9 (Non-Functional Requirements): The quality attributes of the system, categorized under ISO/IEC 25010 standards and associated with measurable criteria, are specified.
- Sections 10 – 12: The business rules, design constraints, and underlying assumptions of the project are detailed.
- Sections 13 – 15: Graphical models, including Use Case Diagrams, Use Case Descriptions, and Activity Diagrams, are incorporated to visualize system processes.
- Section 16 (External Interface Requirements): Expectations for user, hardware, software, and communication interfaces are established.
- Section 17 (Data Requirements): The structure and management of data within the system are outlined.
- Section 18 – 20: Alternative solutions, feasibility studies (Technical, Economic, Operational, and Schedule), and a comprehensive Risk Register are presented.
- Section 21 (References): All external documents and standards cited in the report are listed in IEEE format.

6. Project Scope

The boundary of the Optical Shop Management System (OSMS) is defined by the digitalization of core retail and clinical operations for Darshana Optical Pvt Ltd. The scope is centered on creating a synchronized environment between the physical branches and a centralized web-based management platform.

The project scope is categorized into the following primary domains:

6.1 Retail and E-commerce Integration

A digital storefront is established to facilitate the display of eyewear products, including frames, lenses, and sunglasses. Features such as product filtering by brand, price, and style are provided to enhance the customer's browsing experience. Secure online payment gateways are integrated to support digital transactions.

6.2 Clinical Appointment Scheduling

Manual appointment booking processes are replaced by an automated scheduling module. Real-time availability of optometrists across different branches is displayed, allowing customers to book eye examinations at their convenience. Automated notifications and reminders are dispatched to reduce no-show rates.

6.3 Multi-branch Inventory Management

A centralized inventory database is maintained to provide real-time visibility of stock levels across all physical locations. Automated alerts are generated when stock reaches re-order levels. The movement of goods between branches is tracked to ensure optimal stock distribution.

6.4 Customer Relationship and Retention

Personalized customer profiles are maintained, capturing purchase history and prescription details. A tiered loyalty program is implemented to reward frequent customers with discounts and exclusive offers. Instant communication is facilitated through an integrated AI-driven chatbot for general inquiries.

6.5 Administrative Reporting

Comprehensive reporting tools are provided for management to analyze sales trends, appointment frequency, and inventory turnover. Data-driven insights are generated to support strategic decision-making and business growth.

7. Stakeholders

The key individuals and groups influenced by the implementation of the Optical Shop Management System (OSMS) are identified in this section. Their roles and primary interests in the system are summarized in the following table:

Stakeholder	**Category**	**Role and Interest**
System Administrator	Internal	Full access to the system is granted. Management of user accounts, system configuration, and data security is performed.
Branch Manager	Internal	Oversight of branch-specific operations is provided. Sales reports, staff performance, and inventory levels are monitored.
Optometrist / Medical Staff	Internal	Eye examination schedules are managed. Patient prescription history is recorded and accessed during consultations.
Inventory Manager	Internal	Stock levels across all branches are tracked. Purchase orders are generated, and stock transfers between locations are coordinated.
Sales Assistant / Cashier	Internal	Daily sales transactions are processed. Customer inquiries are handled, and loyalty points are updated at the point of sale.
Customers	External	The online store is accessed for browsing and purchasing eyewear. Appointments are booked, and order statuses are tracked through the web portal.

Darshana Optical Management	External/Owner	High-level business analytics are reviewed. Strategic decisions are made based on the sales and growth data generated by the system.
Third-party Logistics Providers	External	Order dispatch information is received. Delivery status updates are integrated back into the OSMS for customer tracking.

Table 02

8. Functional Requirements

The fundamental services and specific behaviors provided by the Optical Shop Management System (OSMS) are defined in this section. Each requirement is uniquely identified and prioritized based on its importance to the core operations of Darshana Optical Pvt Ltd.

ID	Description	Priority	Source	Status

FR-001	User registration and authentication are provided for customers and staff using secure credentials.	High	System Admin	Draft
FR-002	Profile management is enabled for users to update personal details and view prescription history.	Medium	Customer	Draft
FR-003	Digital product catalogs are maintained, allowing items to be filtered by brand, style, and price.	High	Branch Manager	Draft
FR-004	A secure shopping cart and checkout process are implemented for online eyewear purchases.	High	Customer	Draft
FR-005	Real-time stock levels are synchronized across all physical branches to ensure inventory accuracy.	High	Inventory Manager	Draft
FR-006	Automated alerts are generated by the system when inventory items reach predefined re-order levels.	Medium	Inventory Manager	Draft
FR-007	Online appointment booking for eye examinations is facilitated through an interactive calendar.	High	Optometrist	Draft
FR-008	Automated appointment reminders are dispatched to customers via email or SMS.	Medium	Branch Manager	Draft

FR-009	Integrated payment gateways are utilized to process credit/debit card transactions securely.	High	Management	Draft
FR-010	Loyalty points are calculated and updated automatically based on customer purchase values.	Low	Management	Draft
FR-011	An AI-driven chatbot is provided to handle general inquiries and guide users through the site.	Low	Customer	Draft
FR-012	Comprehensive sales and inventory reports are generated for management review.	High	Management	Draft
FR-013	Prescription data is recorded and stored securely after each clinical eye examination.	High	Optometrist	Draft
FR-014	Order tracking status is updated in real-time as products move through the delivery process.	Medium	Customer	Draft

Table 03

9. Non-Functional Requirements

The quality attributes and operational constraints of the Optical Shop Management System (OSMS) are specified in this section. These requirements ensure that the system is robust, user-friendly, and secure.

Category	Requirement ID	Description	Measurable Acceptance Criterion
Performance	**NFR-001**	System responsiveness is maintained under standard operational loads.	95% of database search queries shall be processed and displayed within 2 seconds.
Reliability	**NFR-002**	Continuous availability of the system is ensured for business operations.	An uptime of 99.9% shall be maintained, excluding scheduled maintenance periods.
Usability	**NFR-003**	The user interface is designed for ease of use by non-technical staff.	New staff members shall be able to process a complete sales transaction within 10 minutes of initial training.

Security	**NFR-004**	Unauthorized access to sensitive customer and prescription data is prevented.	All data transmissions shall be encrypted using SSL/TLS, and passwords shall be hashed using Bcrypt.
Scalability	**NFR-005**	The system architecture is designed to handle business growth.	The system shall support a 50% increase in concurrent users and transactions without a degradation in response time.
Maintainability	**NFR-006**	The codebase is structured to allow for efficient updates and bug fixes.	A modular architecture is utilized, ensuring that 90% of minor bug fixes are implemented and deployed within 24 hours.
Portability	**NFR-007**	Cross-platform compatibility is provided for various user environments.	The web application shall be fully functional across the latest versions of Google Chrome, Mozilla Firefox, and Safari.

Table 04

10. Business Rules

The operational policies and decision-making constraints of Darshana Optical Pvt Ltd are defined in this section. These rules are enforced by the system to ensure business integrity and consistency across all branches.

10.1 Appointment Management Rules

- **Booking Lead Time:** Eye examination appointments are allowed to be scheduled only if they are made at least 24 hours in advance.
- **Cancellation Policy:** Cancellations of appointments are permitted without penalty only if they are performed at least 12 hours prior to the scheduled time.
- **Optometrist Allocation:** Only one patient is permitted to be assigned to a specific optometrist within a single 20-minute time slot.

10.2 Inventory and Sales Rules

- **Stock Re-ordering:** Automated re-order requests are triggered only when the physical stock level of a specific SKU falls below the safety stock threshold defined for each branch.
- **Discount Application:** Seasonal or promotional discounts are applied to the final invoice only if the items are not already marked as "Clearance" stock.
- **Prescription Validity:** Orders for prescription lenses are processed only if the associated eye test result is dated within the last 24 months.

10.3 Loyalty and Customer Rules

- **Point Accrual:** Loyalty points are earned at a rate of 1 point for every 100 LKR spent on frames and accessories; however, points are not accrued for clinical eye test fees.
- **Point Redemption:** Redemption of loyalty points is allowed only when the customer's total point balance exceeds 500 points.
- **Return Policy:** Returns on eyewear frames are accepted only within 7 days of purchase, provided the items are in original condition and accompanied by a valid receipt.

10.4 User Access Rules

- **Managerial Override:** Inventory adjustments and manual price changes are permitted to be performed only by users with "Branch Manager" or "System Admin" clearance.
- **Session Security:** User sessions are automatically terminated after 30 minutes of inactivity to prevent unauthorized access in a retail environment.

11. Constraints

The design and implementation of the Optical Shop Management System (OSMS) are governed by several limiting factors. These constraints are categorized into technical, regulatory, and operational boundaries as follows:

11.1 Technical Constraints

- **Technology Stack:** The development of the system is restricted to the **MERN stack** (MongoDB, Express.js, React, and Node.js) as per the architectural requirements.
- **Web Browser Compatibility:** Full functionality of the administrative and customer portals is required to be maintained only on the latest versions of **Google Chrome, Mozilla Firefox, and Microsoft Edge**.
- **Hosting Environment:** The application is required to be deployed on a cloud-based infrastructure (e.g., AWS or Azure) to ensure high availability and scalability.
- **Connectivity:** A continuous internet connection is necessary for real-time inventory synchronization; offline processing of sales is not supported in the initial version.

11.2 Regulatory and Legal Constraints

- **Data Privacy:** All personal and medical data (prescriptions) are required to be handled in compliance with the **Personal Data Protection Act (PDPA)** of Sri Lanka.
- **Financial Standards:** Integration with payment gateways is permitted only through providers that are certified with **PCI DSS** (Payment Card Industry Data Security Standard).
- **Audit Trails:** Permanent logs of all financial transactions and stock adjustments are required to be maintained for a minimum period of seven years for auditing purposes.

11.3 Operational and Design Constraints

- **Language Support:** The initial user interface is restricted to the **English language** only; multi-language support (Sinhala/Tamil) is excluded from the current development phase.
- **User Interface Consistency:** The design of the system is required to strictly follow the corporate branding guidelines of Darshana Optical Pvt Ltd, including specific color palettes and typography.
- **Budgetary and Schedule Limits:** The development and deployment of the core modules are required to be completed within the allocated project budget and the six-month delivery schedule.

12. Assumptions

The development and deployment of the Optical Shop Management System (OSMS) are based on the following assumptions. These factors are assumed to be true for the successful execution of the project:

12.1 Technical and Infrastructure Assumptions

- **Hardware Availability:** It is assumed that all physical branches of Darshana Optical Pvt Ltd are equipped with the necessary computing hardware (Desktop PCs or Tablets) and stable high-speed internet connectivity.
- **Server Performance:** The cloud hosting environment (e.g., AWS/Azure) is assumed to provide sufficient resources to maintain the performance metrics defined in the Non-Functional Requirements.
- **Third-party APIs:** The reliability and continuous uptime of external services, such as payment gateways (e.g., PayHere) and SMS service providers, are assumed throughout the operational period.

12.2 Data and User Assumptions

- **Data Accuracy:** All existing manual records, including inventory counts and customer prescription history, are assumed to be accurate and verified before being migrated to the new digital database.
- **User Competency:** It is assumed that the staff members of Darshana Optical possess basic computer literacy and will undergo the scheduled training programs to operate the new system effectively.
- **Customer Access:** The majority of the target customer base is assumed to have access to smartphones or computers with internet browsing capabilities to utilize the online store and booking features.

12.3 Organizational and Legal Assumptions

- **Stakeholder Availability:** Timely feedback and approvals from the management of Darshana Optical Pvt Ltd are assumed to be provided during the requirement validation and UAT (User Acceptance Testing) phases.
- **Regulatory Stability:** No major changes to the national data protection laws (PDPA) or financial regulations are assumed to occur during the six-month development lifecycle that would necessitate significant architectural changes.

13. Use Case Diagrams

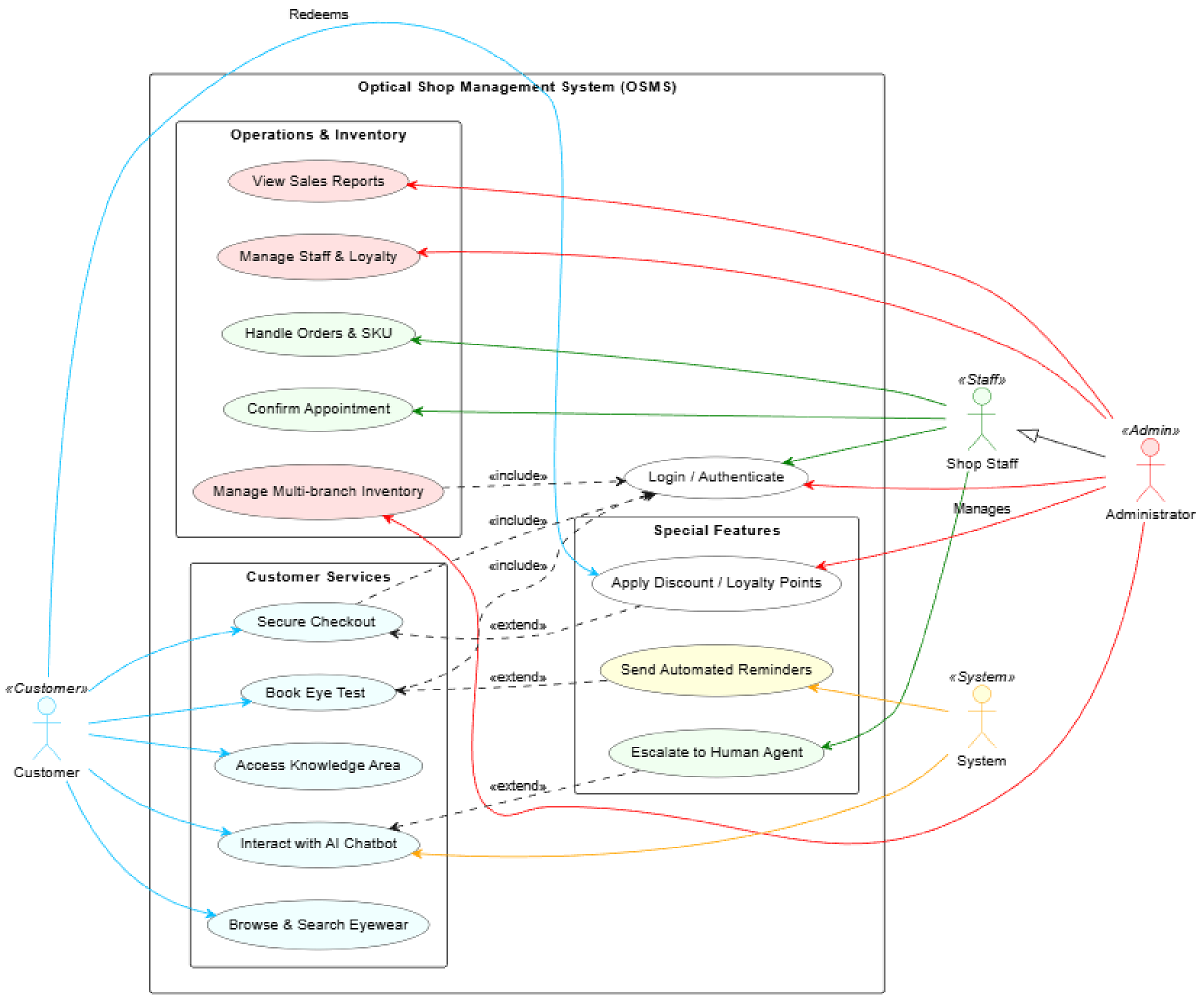

Figure 01

The functional interactions between the identified actors and the Optical Shop Management System (OSMS) are visually represented in this section. The diagram illustrates how different user roles engage with various system modules to perform business operations.

13.1 Identified Actors The system's functionalities are triggered by four primary actors:

- **Customer:** Interacts with the system for browsing, booking eye tests, and making secure purchases.
- **Shop Staff:** Responsible for clinical tasks such as confirming appointments and handling order-related operations.
- **Administrator:** Granted full control to manage staff, loyalty programs, multi-branch inventory, and view analytical sales reports.

- **System:** An automated actor involved in sending reminders and managing background processes.

13.2 Use Case Relationships The diagram incorporates standard UML relationships to define logical dependencies:

- **«include»:** Core processes such as *Manage Multi-branch Inventory* and *Secure Checkout* are shown to necessarily include the *Login/Authenticate* use case to ensure security.
- **«extend»:** Optional or conditional flows are represented, such as *Apply Discount/Loyalty Points* extending the checkout process, and *Escalate to Human Agent* extending the chatbot interaction.

13.3 Functional Grouping As depicted in the diagram, the system's capabilities are organized into three distinct logical boundaries:

- **Operations & Inventory:** Covers backend management tasks restricted to internal staff and administrators.
- **Customer Services:** Encompasses all frontend functionalities accessible to the customer.
- **Special Features:** Includes auxiliary automated services such as AI-driven support and automated notification systems.

14. Use Case Descriptions

Detailed descriptions for the primary use cases identified in the previous section are provided here. Each description outlines the preconditions, post-conditions, and the step-by-step interaction between the actor and the system.

14.1 Use Case: Book Eye Test (UC-03)

Field	**Description**
Use Case ID	UC-03
Use Case Name	Book Eye Test
Actors	Customer, System

Description	The process of selecting a branch, an optometrist, and a time slot for a clinical eye examination is described.
Pre-conditions	1. The customer is required to be logged into a valid OSMS account. 2. Real-time availability of time slots is required to be active in the database.
Post-conditions	1. The selected time slot is marked as "Unavailable" in the system. 2. An automated confirmation is dispatched to the customer.

Main Success Scenario (Flow of Events):

1. The "Book Appointment" option is selected by the Customer from the dashboard.
2. A list of available Darshana Optical branches is displayed by the system.
3. The preferred branch and optometrist are chosen by the Customer.
4. Available time slots are retrieved from the database and presented in a calendar interface.
5. A specific time slot is selected, and necessary contact details are provided by the Customer.
6. The booking request is submitted for processing.
7. The appointment is confirmed by the system, and a unique booking ID is generated.
8. An automated reminder is scheduled to be sent as the appointment date approaches.

Alternative Flows:

- **A1: Slot Already Booked:** If the selected slot is taken by another user simultaneously, an error message is displayed, and the Customer is prompted to select a different time.
- **A2: Invalid Data:** If mandatory fields (e.g., phone number) are left blank, the submission is rejected until the required information is provided.

14.2 Use Case: Manage Multi-branch Inventory (UC-05)

Field	Description
Use Case ID	UC-05
Use Case Name	Manage Multi-branch Inventory
Actors	Administrator, Inventory Manager
Description	Stock levels for frames and lenses are tracked and updated across all physical branch locations.
Pre-conditions	Administrative or Managerial access rights are required to be verified by the system.
Post-conditions	Centralized stock counts are updated, and low-stock alerts are triggered if thresholds are reached

15. Activity Diagrams

The dynamic flow of activities within the Optical Shop Management System (OSMS) is illustrated in this section. The logical sequences, decision points, and concurrent activities for key business processes are captured.

15.1 Overall System Process Flow

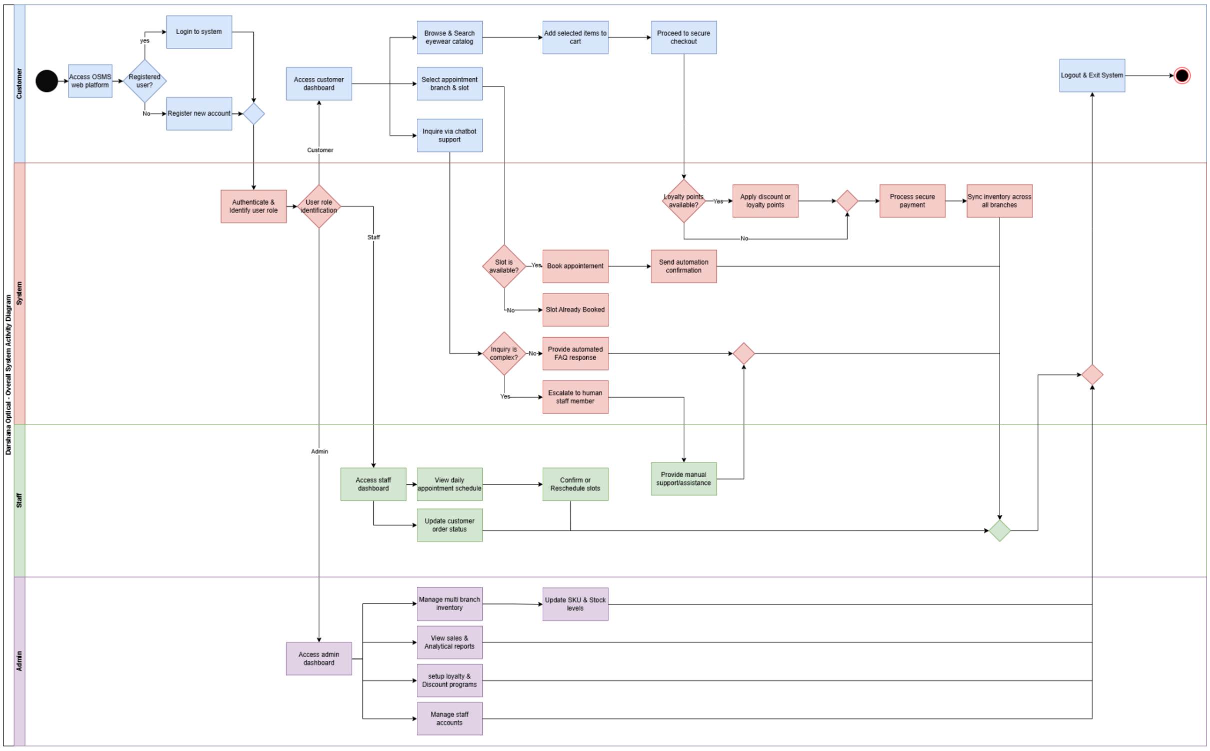

Figure 02

A comprehensive view of the system's operational flow is provided in the integrated activity diagram. The interactions between the Customer, Staff, and Administrator are partitioned using swimlanes to define responsibilities.

- **Customer Path:** The process begins with the access of the web platform, followed by authentication and navigation to the product catalog or appointment dashboard.
- **Staff Path:** Appointment schedules are reviewed and confirmed, and customer order statuses are updated based on branch operations.
- **Administrator Path:** High-level activities including multi-branch inventory management and analytical report generation are executed.

15.2 Detailed Activity: Online Appointment Booking

The specific logic for securing an eye examination is described as follows:

1. The appointment module is initiated by the Customer.
2. Branch and Optometrist selections are performed.

3. **Decision Node:** The system checks if the selected slot is available in the database.
4. If available, the slot is reserved, and a confirmation is generated.
5. If unavailable, an error message is displayed, and the flow returns to the slot selection stage.
6. The process is concluded with the dispatch of an automated reminder

16. External Interface Requirements

The requirements for the interfaces between the Optical Shop Management System (OSMS) and external entities, including users, hardware devices, and third-party software services, are specified in this section.

16.1 User Interface Requirements

- **Responsiveness:** A modern and responsive user interface is developed using React.js to ensure seamless operation across desktops, tablets, and smartphones.
- **Ease of Navigation:** The UI design is focused on intuitive navigation to accommodate both tech-savvy audiences and older customers browsing the catalog or booking appointments.
- **Accessibility Standards:** Compliance with the Web Content Accessibility Guidelines (WCAG) is maintained to ensure that the "Knowledge Area" and store functions are accessible to all users.
- **Consistency:** A clean and professional aesthetic is maintained throughout the system to reflect the corporate branding of Darshana Optical Pvt Ltd.

16.2 Hardware Interface Requirements

- **Computing Devices:** The system is accessed via any standard computing device equipped with an internet connection and a web browser.
- **Peripheral Support:** Standard retail peripherals, including barcode scanners for inventory tracking and receipt printers for in-store order processing, are supported for shop staff.
- **Camera Integration:** Web camera access is utilized if virtual try-on features are implemented for customers.

16.3 Software Interface Requirements

- **Database Management:** The backend (Node.js/Express.js) interfaces with MongoDB for the storage and retrieval of customer profiles, inventory levels, and clinical records.
- **Messaging Services:** Integration with third-party SMS and Email gateways, such as Twilio or Nodemailer, is performed to dispatch automated reminders and booking confirmations.
- **Operating Systems:** The application is developed to be platform-independent, running on Windows, macOS, and Linux via modern web browsers.

16.4 Communications Interface Requirements

- **Communication Protocol:** All data transmissions between the React frontend and the Node.js backend are conducted via secure HTTPS protocol to ensure encryption of sensitive data.
- **Architectural Style:** Communication is facilitated through RESTful APIs to manage the exchange of data between the client and the server.
- **Data Format:** JSON (JavaScript Object Notation) is utilized as the primary data format for all inter-component communications.
- **Network Standards:** Standard TCP/IP network protocols are followed to ensure stable connectivity between branch locations and the centralized cloud database

17. Data Requirements

The organization, storage, and management of information within the Optical Shop Management System (OSMS) are defined in this section. A structured approach to data handling is necessitated to ensure consistency across all branches of Darshana Optical Pvt Ltd.

17.1 Database Management System

- **Non-relational Structure:** A schema-less but structured approach is utilized within **MongoDB** to accommodate diverse product specifications and complex customer prescription histories.
- **Centralized Storage:** All operational data, including inventory levels and appointment schedules, is maintained in a centralized cloud-based database to facilitate real-time synchronization.

17.2 Data Entities and Relationships

The logical structure of the database is illustrated through an **Entity Relationship (ER) Diagram**. The following key entities are identified and managed within the system:

- **User Entity:** Data related to Customers, Staff, and Administrators, including encrypted authentication credentials, is stored.
- **Product Entity:** Information regarding eyewear frames, lenses, and accessories is categorized by unique **SKUs**, brands, and styles.
- **Appointment Entity:** Records of eye test bookings, including timestamps, assigned optometrists, and branch locations, are maintained.
- **Prescription Entity:** Sensitive clinical data from eye examinations is securely recorded and linked to specific customer profiles.
- **Order Entity:** Transactional data, including purchased items, payment status, and delivery tracking information, is captured.

17.3 Data Integrity and Retention

- **Synchronization:** Real-time updates are performed during every sale or stock transfer to prevent data conflicts between different branches.
- **Security:** Sensitive data, such as eye prescriptions and personal contact details, is protected using encryption protocols (SSL/TLS) during transmission and storage.
- **Backup and Recovery:** Automated backups of the entire database are scheduled regularly to ensure data recoverability in the event of a system failure.
- **Archiving:** Historical sales and clinical records are retained for a minimum of seven years to comply with auditing and regulatory requirements.

18. Alternative Solutions Considered

The evaluation of various strategic and technical approaches for the digitalization of Darshana Optical Pvt Ltd is detailed in this section. Several alternatives were considered before the current web-based MERN stack solution was finalized.

18.1 Maintaining the Existing Manual System

The continuation of the current paper-based record-keeping and physical appointment scheduling was evaluated.

- **Reason for Rejection:** This approach was rejected due to inherent inefficiencies, including manual tracking errors, lack of real-time inventory visibility across branches, and the inability to reach a tech-savvy customer base.

18.2 Development of a Native Desktop Application

The implementation of a standalone desktop software installed at each physical branch was considered.

- **Reason for Rejection:** High maintenance costs and difficulties in real-time data synchronization between multiple geographical locations were identified as primary drawbacks. Furthermore, the requirement to provide a digital storefront for external customers could not be fulfilled through a traditional desktop environment.

18.3 Utilization of Off-the-shelf Retail Software

The purchase of a generic, pre-built retail management system was researched.

- **Reason for Rejection:** While cost-effective initially, generic systems were found to lack the specific clinical modules required for eye examinations and prescription history management unique to an optical shop. Customization of such software to include the AI-driven chatbot and specific loyalty programs was deemed technically restrictive.

18.4 Alternative Technology Stacks (e.g., PHP/MySQL)

Traditional web development stacks such as LAMP (Linux, Apache, MySQL, PHP) were assessed.

- **Reason for Rejection:** Although reliable, these stacks were considered less efficient than the **MERN stack** for handling real-time, high-concurrency data updates. The single-page application (SPA) capabilities of **React.js** were preferred to ensure a modern, fluid user experience that meets the expectations of the target audience.

19. Feasibility Study

The viability of the Optical Shop Management System (OSMS) is assessed across four key dimensions to ensure that the project is technically achievable, economically sound, operationally compatible, and deliverable within the required timeframe.

19.1 Technical Feasibility

The technical feasibility of the project is confirmed based on the following factors:

- **Technology Stack:** The selection of the **MERN stack** (MongoDB, Express.js, React, Node.js) is supported by the availability of extensive documentation and robust library support.
- **Development Expertise:** The required skills for implementing RESTful APIs, secure authentication (JWT), and responsive web design are possessed by the development team.
- **Infrastructure:** Cloud-based hosting environments (e.g., AWS or Azure) are readily available to meet the real-time synchronization and scalability requirements of Darshana Optical Pvt Ltd.
- **Integration:** Proven methods for integrating third-party services, such as payment gateways and SMS notifications, are established and widely used in similar retail applications.

19.2 Economic Feasibility

The economic impact and cost-benefit analysis are evaluated as follows:

- **Development Costs:** The utilization of open-source technologies (Node.js, MongoDB) significantly reduces initial software licensing expenses.
- **Long-term Savings:** Operational costs are reduced through the automation of manual inventory tracking and the minimization of errors in appointment scheduling.
- **Revenue Growth:** A broader customer base is accessed through the implementation of the online store and digital marketing capabilities, leading to increased sales.

- **Return on Investment (ROI):** The initial investment is projected to be recovered through enhanced efficiency and customer retention achieved via the loyalty program.

19.3 Operational Feasibility

The compatibility of the system with the existing business environment is assessed:

- **Workflow Alignment:** The system is designed to streamline existing processes, such as eye test bookings and stock management, without disrupting the core clinical services of the shop.
- **User Acceptance:** A user-friendly interface is provided to ensure that staff with moderate technical skills can operate the system with minimal training.
- **Management Support:** Strong commitment is shown by the management of Darshana Optical to transition from manual to digital operations to overcome current business limitations.

19.4 Schedule Feasibility

The ability to deliver the system within the stipulated timeframe is analyzed:

- **Project Timeline:** A structured six-month development schedule is established, covering requirement elicitation, design, development, and testing.
- **Phased Development:** The project is divided into manageable modules (e.g., User Management, Online Store, Inventory), allowing for incremental progress and timely reviews.
- **Resource Allocation:** Sufficient time and personnel are allocated to handle critical path activities, ensuring that the project deadlines are met.

20. Risk Analysis

Potential threats to the successful completion and operation of the Optical Shop Management System (OSMS) are identified and evaluated in this section. Each risk is documented in a formal Risk Register to establish proactive mitigation strategies.

Risk Register Table

Risk ID	Description	Category	Likelihood	Impact	Risk Level	Mitigation Strategy
RSK-001	Data loss occurs during the migration from manual records to MongoDB.	Technical	Low	High	Medium	Incremental data migration is performed, and comprehensive backups are maintained prior to the transition.
RSK-002	Delays are encountered in integrating third-party payment gateways.	Schedule	Medium	High	High	Integration tasks are initiated early in the development phase, and alternative providers are pre-identified.

RSK-003	Key development personnel become unavailable during critical phases.	Resource	Low	Medium	Medium	Thorough code documentation is maintained, and knowledge-sharing sessions are conducted regularly.
RSK-004	Changes to functional requirements are requested after the sign-off period.	Requirements	Medium	High	High	A formal change control process is implemented, and the impact on the schedule is communicated to the client.
RSK-005	Unauthorized access to sensitive eye prescription data is attempted.	Technical	Low	High	Medium	Multi-factor authentication and high-level encryption (SSL/TLS) are strictly enforced.
RSK-006	The system is rejected by staff due to perceived complexity.	Organisational	Medium	Medium	Medium	Comprehensive user training sessions are conducted, and a user-centric UI design is prioritized.

Table 06

21. References

[1] "Optical Shop Management System for Darshana Optical Pvt Ltd," Project Proposal, Darshana Optical Pvt Ltd, Sri Lanka, 2026.

[2] *IEEE Recommended Practice for Software Requirements Specifications*, IEEE Std 830-1998, 1998.

Client Requirement Sign-off Page

Client Representative Sign-off

Name: ____________________________________

Title: ____________________________________

Organisation: ____________________________________

Date: ____________________________________

Signature: ____________________________________

Project Supervisor Sign-off

Name: ____________________________________

Date: ____________________________________

Signature: ____________________________________

Note: Any changes to the approved requirements after this sign-off must be submitted as a formal Change Request and re-approved by both parties before implementation.